

MOU PAUL

Engineering leader focused on reliable, large-scale infrastructure. Proven success architecting token pipelines at trillion-scale, leading high-stakes integrity and anti-spam initiatives, and delivering \$10M+ in infra efficiencies.

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paul.mou.dev

Technical Skills

Language Python Golang Typescript/Javascript Rust
Platform Kubernetes Docker Spark BigQuery BigTable Node.js gRPC
Infra Kafka Consul Vault Nginx Redis Sqlite
Library Ray Pytorch Rocksdb Electron Next.js React

Selected Projects

Tiny World Model
<https://paul.mou.dev/posts/2026-08-16-twm/>
Building Flow Matching Models from Scratch
<https://paul.mou.dev/posts/2026-03-10-flow/>

Professional Experiences

- Member of Technical Staff @ Microsoft AI, Redmond 2024/12-Present
- Architected petabyte-scale Spark/Ray data pipelines to process trillions of tokens from 3B+ unstructured PDFs for LLM pre-training, utilizing ablation studies to refine data-mixture strategies
 - Engineered automated filtering and deduplication workflows using vector embeddings and LLM-as-a-Judge to curate high-fidelity training sets from raw, unstructured data
- Software Engineer @ Roblox, San Mateo 2022/05-2024/12
- Led the evolution of Roblox's analytics ingestion infrastructure, scaling to trillions of events a day while delivering \$10M+ in annual cloud savings with datalake schematization
 - Promoted DevOps best practices via code-based dashboard and alert generation, runbook documentation, capacity planning, and incident postmortems
- Software Engineer @ Twitter, San Francisco 2018/06-2022/04
- Scaled smyte's anti-spam infrastructure 10x post-acquisition through rearchitecture and integration with Twitter infrastructure
 - Hardened data privacy for GDPR compliance by designing a personal data anonymizer system for the anti-spam infrastructure
- Software Engineer @ Smyte, San Francisco 2015/11-2018/06
- Core member of four engineers building and scaling Smyte's anti-spam engine protecting hundreds of millions of users
 - Developed and maintained automation and monitoring for multiple kubernetes cluster with thousands of pods
 - Contributed to high impact initiatives such as GDPR compliance and building first machine learning stack for abusive image detection

Educations

Udacity	Stanford University	University of Waterloo
Self-Driving Car Engineer I Certificate	Computer Science Master (withdrew)	System Design Bachelor